Assignment 1: Branch Predictor Implementation

Course 6385

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1 What was implemented

Two classes in `my_predictor.h` implement the same hybrid. `pm_predictor` is the Appendix B worksheet: 16 bimodal counters, four global sets, and a 4-bit history. It exists so the worked example can be checked cell for cell. `scaled_pm_predictor` is the predictor `predict.cc` runs on the traces. It uses the Pentium M table sizes and the same selection rule. The assignment allows two ways instead of the Pentium M's four.

A conditional branch in `scaled_pm_predictor` is predicted as follows.

- The bimodal table has 4096 two-bit saturating counters. The index is `PC[11:0]`. The prediction is the counter's high bit.
- A 15-bit history register records the outcome of each conditional branch. It is shifted after the branch resolves, and the new taken or not-taken bit is inserted. This is the stand-in for the path information register, which the assignment does not require.
- `PC[16:2]` is XORed with those 15 history bits, the same split as Appendix B at a wider width. Bits `[14:6]` of the result select one of 512 sets. Bits `[5:0]` are the tag.
- Each set has two ways. A way hits when it is valid and the tag matches. On a hit the global counter's high bit is the prediction. On a miss the bimodal prediction is used.
- Both counters are updated. A miss fills an invalid way, or replaces the least-recently-used way if both are valid. The chosen way becomes most recently used.

Counters saturate at 0 and 3 and start at 0. The worksheet class keeps the Appendix B initial state: bimodal counter i starts at i mod 4, and the global ways start with the valid bits clear and the counter and LRU values shown in that example.

2 What was left out

Loop prediction, the branch target buffer, and the path information register are not implemented. Unconditional branches are predicted taken and do not update the tables. Target addresses are not predicted; the simulator still reports direction mispredictions per kilo-instruction, which is the metric for this assignment.

3 Check against Appendix B

Appendix B gives 22 conditional branches (`b1` at `0x44`, `b2` at `0x6C`, `b3` at `0x90`) and the expected state of both tables. Those tables are stored as `hybrid.csv`, `local.csv`, and `global.csv`. The program `appendix_b_check` feeds each program counter and outcome to `pm_predictor` and compares the history register, the XOR index, the 16 bimodal counters, and every global valid bit, tag, counter, and LRU bit.

The run on `pm_predictor` reports 22 of 22 rows matching. `scaled_pm_predictor` is not compared to those tables. Its index width is not the 4-bit index in the example.

4 Trace results

The predictor was built with `make` in `src` and run on the 20 CBP2 traces. Each trace is 100 million instructions. MPKI is $1000 \times (\textit{direction mispredictions})/10^8$, counted only for conditional branches. The always-taken column is the same traces run with the original stub, which predicted every conditional branch taken.

Trace	Always taken	Hybrid
164.gzip	70.025	13.868
175.vpr	85.312	17.028
176.gcc	86.071	20.021
181.mcf	102.497	21.085
186.crafty	48.651	10.957
197.parser	78.421	12.861
201.compress	76.983	9.230
202.jess	92.343	2.461
205.raytrace	109.506	3.975
209.db	91.451	5.583
213.javac	109.644	3.328
222.mpegaudio	110.339	3.273
227.mtrt	112.022	3.898
228.jack	77.958	5.793
252.eon	24.226	4.074
253.perlbmk	66.833	8.127
254.gap	75.950	6.829
255.vortex	70.524	2.100
256.bzip2	178.257	0.191
300.twolf	59.221	24.558
Average	86.311	8.962

Table 1: Direction mispredictions per kilo-instruction. The hybrid column is `scaled_pm_predictor`.

The hybrid predictor is lower than always-taken on every trace. The average falls from 86.311 to 8.962 MPKI. bzip2, vortex, and jess fall the furthest. gcc, mcf, and twolf stay above 20. Those traces have more distinct branches than a 512-set, 2-way table can separate, so some global hits still come from a counter trained by a different branch.

5 Conclusion

`pm_predictor` is the Appendix B hybrid and matches the worked example on all 22 rows. The trace predictor, `scaled_pm_predictor`, is that same machine at Pentium M sizes: a 4096-entry bimodal table, a 15-bit outcome history, and a 512-set 2-way tagged global table. A global hit overrides the bimodal prediction. On the CBP2 traces the average falls from 86.311 to 8.962 MPKI.